

Implementation Summary

Step : 1 Clone The Repo From Master Branch

Clone the project repository from the master branch to your local machine. This ensures you are working with the latest stable version of the codebase before starting the implementation.

```
git clone <repository-url>
```

```
git checkout master
```

Step 2: Install Dependencies

Run `npm install` to download and install all required project dependencies specified in the `package.json` file. This prepares the development environment and ensures the application can run correctly.

```
npm install
```

Step 3: Start the Application

Run `npm start` to launch the application in the local development environment. This starts the server and allows you to access and test the application.

The server and client will start automatically and run concurrently, as this configuration is defined in the root `package.json` file.

```
npm start
```

Step 4: Open the Application

After the application starts successfully, open the local URL displayed in the terminal (e.g., `http://localhost:3000`) in your browser to access and test the application.

Step 5: Access the Lobby Page

Once the application is loaded, users are redirected to the Lobby page. The page establishes a Socket.io connection with the server and displays all active rooms available for collaboration in real time.

Step 6: Create a New Room

Enter a username and optionally provide a room name. Click **Create Room** to generate a unique 6-character room code. The server creates the room, stores the user information, and redirects the creator to the room page.

Step 7: Join an Existing Room

To join an existing room, enter a username and the room code, then click **Join Room**. The server validates the room code and, upon success, adds the user to the room and updates all connected participants.

Step 8: Collaborate Inside the Room

Inside the room, users can:

- View active room members.
- Send and receive real-time chat messages.
- Trigger activity events such as File Upload, Page View, and Button Click.
- Receive live notifications when users join or leave the room.

Step 9: Monitor Real-Time Activity

Navigate to the Dashboard page to view live system statistics, including active rooms, connected users, and event history. All room activities are streamed in real time through Socket.io.

Step 10: Leave the Room

Users can leave the room by clicking the **Leave** button. The server removes the user from the room and notifies the remaining participants. Empty rooms are automatically removed from the system.

Step 11: Verify Real-Time Synchronization

Open multiple browser tabs or devices and join the same room. Verify that chat messages, member updates, and activity events are synchronized instantly across all connected clients.